DAY 1 Data Literacy In A Data And AI Driven World

Tuesday, 3 November 2026 07:23

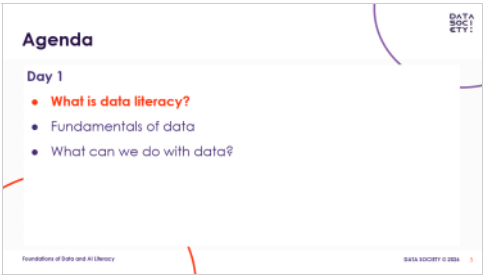
DAY 1 Foundati...

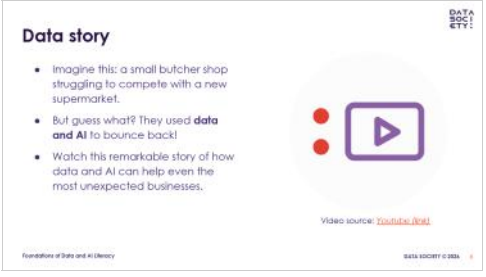












May I ask how AI came into the picture of the butcher shop? Is it to identify camera feed only or is it to support the business aspect only? (He talked a lot about Big Data, but not haw AI was used for processing och advertisement and so forth)

-- In this case AI was used to process the data. Understanding trends. The counting of camera tracking, probably AI in the apparition of them and that tracking.
-- He can also use it from marketing, but we do not know.



Defining data literacy

- Data is a valuable asset for businesses, but data literacy is what brings it to life.
- **Data literacy** is the ability to:
  - **Read**
  - **Write**
  - **Communicate with data** in context — in both work and life through development of one's mindset, language and skills.



What is required to be data literate?

MINDSET	LANGUAGE	SKILLS
Being open, willing & curious. Seeing the world through data glasses.	Business, Data + Analytics Acumen. "**Speaking Data**" as a shared language.	Thinking critically. Engaging others. Applying data constructively.

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Developing mindset

- Cultivating a **data-driven mindset** requires a **shift** from relying on intuition and gut feelings to embracing the power of data.
- To achieve this, it is vital to:
  - Understand the data and steps performed within data analysis.
  - Approach data and analysis with a curious and questioning mind.



Enhancing skills

- Data skills is not just about using data, tools or technology.
- To truly unlock the power of data, begin by cultivating essential **non-technical** skills:
  - **Critical thinking** to ask meaningful questions to the data
  - **Collaboration skills** to work effectively with others and engaging team members, stakeholders
  - Ability to take **constructive and ethical action** when using data



Need for a shared language

- We interact with people from various backgrounds:
  - Industry sector experience
  - Business domain experience
  - Data vs. analytics backgrounds
  - Global vs. local
  - Veterans vs. rookies
- While a variety of perspectives is beneficial, it can also lead to communication challenges.
- **Creating and nurturing a shared language is fundamental.**



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Chat question

How do you think becoming data literate and having a shared language around data would impact the work you do?



ANSWERS:
- Help decision making
- Convey risk
- Efficiency

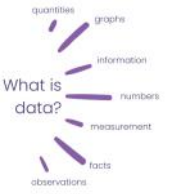
This assumes that there are others that also speak this language. Who would this be and what do they do?

Agenda

- What is data literacy?
- **Fundamentals of data**
- What can we do with data?

What is data?

- **Data** is everywhere. So much so that there's possibly no field that doesn't deal with data.
- It is **information, facts, or numbers** collected to be examined, considered, and used to help decision-making.
- The data universe continues to grow relentlessly.



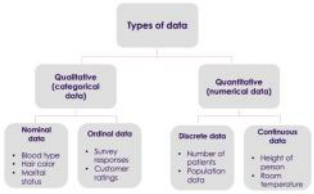
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The current figure for the data created each day is 402.74 million terabytes.
181 zettabytes of data were generated in 2025. The estimate for 2026 is 221 zettabytes.

Every field is generating data, although not much data or handling the data, but data is made everywhere and uploaded from anywhere. Note, opinions aren't facts, but they are still data.

Types of data



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Certain types of data work better for different types of analysis.
Quantitative is always a number data, but not all numbers are numerical data. Zip-codes or telephone numbers are numbers, but not numerical data.
We are not doing any mathematical changes to the numbers.

Nominal Data - The order doesn't matter.
Ordinal Data - The order does matter or has an impact.
Discrete Data - There is a limit to how much you can break the data down to. There is a limited amount of data points. Think of it as stairs.
Continuous Data - Can be anything on the scale and thus has infinite amounts of datapoints. Think of it as a ramp.

Types and formats might sound similar. Types is what type of data, format is how it is structured

Data formats



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Structured: What most people think about as data. It is tight and orderly, everything has a place and there are guidelines for what data goes where. This is what computers and people like as the structures are simple to read and understand.

Unstructured: "The wild west" Data is being produced without a parallel between the data points. Any form of data can also be produced. This also includes ex. A smart fridge that creates data when talking to the home that is in general data that might be anything or any format.

Semi/Quasi -structured: It has some structure, but within ex. the cells it is not determined how the data must be structured. So for the book code, some books will contain an author, some not, some will have page numbers, some not.





b: semi structured. Header, to, from, date and body will always be determined and structured, but what is inside the body is unstructured. It can contain any information and any type of data.

B or C: Within some given fields there might be enough structures on how the data and text must look that it can be called a semi-structured data. But in general it is unstructured, as any type of data might be anywhere at any point of time. And if an image is included, sometimes text is below it, sometimes not. Sometimes it is '*Fig X*' sometimes it contains a '*author*'

Sources of data



e.g. Shared email lists from other companies

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Did your company make the data, gather it themselves on our tools



NOTE: this will be the definition for this class. It doesn't always need to be confined by these limitations, but this definition is not wrong.

Big data is more than a large volume of data. But other than high volume, it needs to flow rapidly aka be updated often. It should also be big in the sence that it contains a lot a factors and types of data that needs to be filtered. The data comes from a lot of different places.

It is thus often a large set of raw information which cannot be processed with general data structuring methods

A social media flow is thus a good example.

Characteristics of big data



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Heterogeneous data

- Big data technology offers incredible opportunities for accelerating discoveries and innovations, but if not used responsibly, it poses significant risks.
- It is important to ensure there is a broad spectrum of data, i.e., different kinds of objects are represented.
- Data that reflects the **a variety of formats, sources, experiences and levels of details** can help fill in the gaps and provide a more rounded and accurate picture.



Image source: Unsplash by Shubham Prakash

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In general the big data is Heterogeneous, but it doesn't have to be big data only bc. It is heterogeneous.

Chat question

Where do you like to grocery shop? And do you like the way they have it set up?



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Answers:

- Costco Yes
- Aldi's and no
- Yes I like the way its set up
- Local market, no
- hypermarket, in an organised manner
- supermarket - sometimes
- On payday, Wegmans, I do
- labelling and section
- Spar Yes
- Local Store - No
- Local vegetable vendor, yes the have it organised and all up front
- Mostly on Hyper market stores.
- I like to shop in the same shop as I know the set up (the structure). I hate when they move it around.

Turning data into action

- Ever noticed how easy it is to find what you need in a massive supermarket?
- That's no accident!
- These stores use data and analytics to design their layouts to offer a smoother shopping experience to their customers.



Watch the video: How Walmart is using data

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scary, however, understandable
a lot of technology and data are using to predict
no
Nice to get curated deals, but not often they are very accurate
Identifying patterns and habits for more customer centric store optimization
No, they do not use technology for sure. small town market.
can we go a bit faster
Yes: In large retailers a must to market and sell products.
not everyone will be able to use this Tech- Especially aged people
but they will lose other sales right
Before typing a thing in the search command i get those items on the page on some online shopping platforms.

Framework for a data-driven approach



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1 Value: Defining the business problem

- Defining the problem is **articulating the business objective** and understanding the **key requirements and constraints.**
- A well-defined problem sets the stage for a successful data project that delivers value to the organization.



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Most important if you are not technical. This will set the stage so that the correct and useful data can be found. We doesn't want to look where the data is the easiest and most comfortable to find.

The streetlight effect

- **The streetlight effect** is a term used to describe a situation where people look for information where is it most convenient.



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The streetlight effect (cont'd)

- To gain a more comprehensive understanding, we should ask ourselves:
  1. What are you searching for?
  2. Where are you searching?
  3. What is your measure of success?
  4. In what ways could your own assumptions influence your data-gathering process?



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We need to look beyond the obvious symptoms and identify the underlying causes of a problem. Even before getting to the analysis we need to:

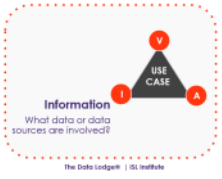
- Continue to ask questions, 5 why
- Understand the problem (it should be described and understood correctly?)

But we can also:

- stop and think. ask for second opinion, review with a stakeholder.
- Critical thinking. End to end search thinking of possibilities.
- Data structure and analyze patterns to generate more data driven insights

2 Information: Finding the right data

- The next step is to identify the data that can help you answer your questions or solve your business problem.
- The data required can be primary data *(collected by yourself or your team or your organization)* or secondary data *(obtained from existing external sources.)*
- It is essential to consider the reliability, validity, and relevance of each data source before using it.



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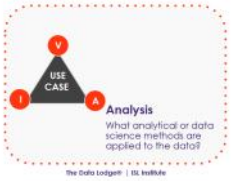
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Should it be internal or external data,
Primary or secondary. Do we need to collect more data to get where we need to be. And how valid will this data be (both if primary or secondary).

3 Analysis

- **Data analysis** focuses on processing and performing statistical analysis on existing datasets.
- Analysts **capture, process, and organize data to uncover actionable insights** for current problems and establish the best way to present this data.



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A insight I snot something you create, it is something you uncover.

Traditional data analytics maturity model



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Model revisited



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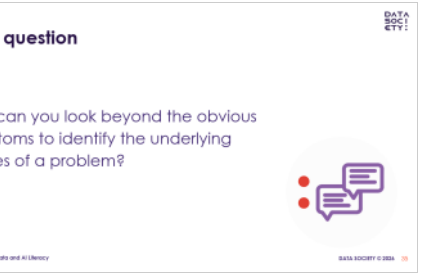
Note that these are accumaltive. You need a earlier stage to do the second stage. You cannnot predict anything without analyzing the past.

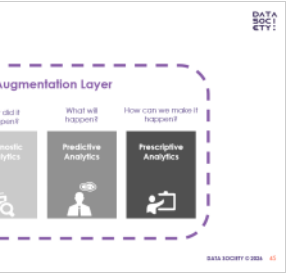
Earlier we said that AI would be the next step, but now all steps have become better via AI

Realm of AI



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question
can you look beyond the obvious
toms to identify the underlying
es of a problem?
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Augmentation Layer
What will happen?
How can we make it happen?
Predictive Analytics
Prescriptive Analytics
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Stage 1: Descriptive analytics

What questions does it answer?	What has happened in the past?
How valuable is it?	Provides some value, but doesn't provide causation or prediction
How labor intensive is it?	Easy to deploy provided you have the right data

Stage 2: Diagnostic analytics

What questions does it answer?	Why did something happen in the past?
How valuable is it?	Provides insights into a particular problem, and can help you identify some root causes for past trends and behaviors
How labor intensive is it?	Requires detailed data, but doesn't have to be overly intensive

Stage 3: Predictive analytics

What questions does it answer?	What is likely to happen?
How valuable is it?	Provides trends / behaviors that are likely to happen
How labor intensive is it?	Requires detailed data, and may require a moderate to high level of computer power, depending on the method and the amount of data

Stage

What qu	
How val	
How lab	

Example: FHA

- The Federal Housing Authority (FHA) forecasts default, repayment, and claim rates using big data analytics.
- They also leverage analytics to anticipate the premiums needed to maintain positive cash flow.
- During the 2008 recession, they were the only subprime mortgage insurance fund that did not require bail out.



Image source: Designed by sumkajanto on freepik.com

Example source: Five Examples of How Federal Agencies Use Big Data | IBM Center for The Business of Government.

Chat question

[Select ALL that apply]
What type of data analytics is demonstrated by the Federal Housing Authority example?

- ❑ Descriptive
- ❑ Diagnostic
- ❑ Predictive
- ❑ Prescriptive



Since they are cumulative, all models are used, but the analytics has been used to get predictive and prescriptive information. Decision were made on these results, so hence it is most likely prescriptive

Summary

Today, we learned about:

- What is data literacy?
- Fundamentals of data
- What can we do with data?

Next session, we will learn about:

- AI and its subfields
- Importance of data management



e 4: Prescriptive analytics



estions does it answer?	What action should I take next?
uable is it?	Provides recommendations for future actions
or intensive is it?	Requires a lot of detailed data, as well as data from other external sources that will impact the model; very labor intensive